Experiment 10 - Experiment to understand array function in Data science.

Aim:

To explore and execute various applications of the pandas library within data science.To conduct an experiment to understand the array function in Data science.

Program:

```
import numpy as np
```

```
array = np.random.randint(1, 100, 9)
```

```
array
```

```
array([[30, 76, 68],
       [70, 39, 58]])
```

```
np.sqrt(array)
```

```
array([7.        , 9.        , 2.44948974, 2.64575131, 3.60555128,
       5.        , 2.82842712, 8.83176087, 3.        ])
```

```
array.ndim
```

```
1
```

```
new_array = array.reshape(3, 3)
```

```
new_array
```

```
array([[49, 81,  6],
       [ 7, 13, 25],
       [ 8, 78,  9]])
```

```
new_array.ndim
```

```
2
```

```
new_array.ravel()
```

```
array([49, 81,  6,  7, 13, 25,  8, 78,  9])
```

```
newm = new_array.reshape(3, 3)
newm
```

```
array([[92, 25, 65],
       [20, 52, 32],
       [ 6, 10, 39]])
```

```
newm[2, 1:3]
```

```
array([10, 39])
```

```
newm[1:2, 1:3]
```

```
array([[13, 25]])
```

```
new_array[0:3, 0:0]
```

```
array([], shape=(3, 0), dtype=int64)
```

```
new_array[0:2, 0:1]
```

```
array([[49],
       [ 7]])
```

```
new_array[0:3, 0:1]
```

```
array([[49],
       [ 7],
       [ 8]])
```

```
new_array[1:3]
```

```
array([[ 7, 13, 25],
       [ 8, 78,  9]])
```

Result:

Thus, an experiment to understand array functions in Data science was conducted.